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RESEARCH ARTICLE

Impact of IBR Location and Parameters on Inter-Area Oscillation Modes in Bulk Power Grids

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ABSTRACT To ensure safe and reliable operations, electric utilities must understand how power grid dynamics are evolving as the existing synchronous machine-dominated systems incorporate increasing amounts of inverter-based resources (IBRs). A pressing concern is understanding if and how the well-known inter-area modes of oscillation will change due to increasing inverter penetration. To address this question, this paper derives an explicit analytical expression to identify the major factors influencing changes in inter-area mode properties when synchronous machines are replaced by IBRs. The IBRs are assumed to implement droop-based grid forming (GFM) control, while the analysis can be extended to other inverter control methods. It is concluded that oscillation mode changes are highly dependent on the grid location where synchronous machines are removed and/or IBRs are added, and the observed changes can be mitigated, to some extent, by tuning two of the GFM inverter control parameters, namely 1) the active power-frequency droop coefficient; and 2) the time constant of the active power measurement low-pass filter. The analytical conclusions are validated using full dynamic simulations of the IEEE 39-bus benchmark system, and the 2031 heavy-winter planning model of the US Western Interconnection. Conclusions from this study will help utilities understand 1) if/which inter-area modes will continue to be of concern in their footprints in an IBR-dominated future, and 2) identify areas where the displacement of synchronous machines may significantly alter the properties of existing modes, necessitating additional analysis during interconnection studies.

INDEX TERMS Inter-area oscillation modes, damping ratio, grid-forming inverters, eigenvalue sensitivity.

I. INTRODUCTION

As synchronous generators are replaced by inverter-based resources (IBRs) to accelerate grid decarbonization, the dynamic behavior of the bulk power grid is changing. Recently, system operators with high IBR deployment have reported increased occurrences of forced oscillations in their footprints. These oscillations may at times interact with inter-area modes and propagate across electrical interconnections [1], [2]. Such experiences have prompted an

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industry-wide interest, in understanding how grid oscillatory characteristics will evolve as the generation mix changes.

It is well known that inter-area oscillatory modes are characterized by how groups of generators in distant parts of a power system exchange energy. This energy exchange manifests as low-frequency oscillations in the 0.1-1 Hz range [3], [4]. Besides frequency, a power system's modes are also characterized by their damping ratio (DR) and shape. The DR indicates how quickly oscillation amplitudes decay, and hence poorly damped modes are a concern for system stability. Mode shapes are defined by two parameters: magnitude and angle. Strong modal participation is visible

at generators with high mode shape magnitudes. Generators with similar mode shape angles oscillate together in phase, whereas those with angles roughly 180° apart oscillate out of phase.

Oscillation mode properties are known to vary with system conditions, and the risk of undamped oscillations is higher when a system is under stress [5], [6], [7]. To formulate effective mitigation actions when damping falls below a threshold, system operators must study mode properties and factors affecting them, and identify generators with strong modal participation. Mode monitoring devices and/or damping controllers are also deployed and configured based on this understanding. Hence, if modal behavior changes significantly, these mitigation actions and monitoring procedures may need reevaluation. Historic evidence suggests that mode properties may change or new modes may emerge if a system undergoes significant changes. For example, following the interconnection of the Alberta region to WI in the 1980s, a ~ 0.25 Hz mode (known as the North-South A mode today) appeared [8].

In recent years, several researchers have explored potential changes in oscillation modes of large interconnections as synchronous machines are displaced by IBRs, using both real and synthetic grid models [9], [10], [11], [12], [13], [14], [15], [16], [17]. These studies reveal that a straightforward correlation does not always exist between the system-wide IBR penetration percentage and observed changes in mode properties. Inverter sizing, location, control mode, and control parameters are some other factors that strongly influence mode behavior. Recent operational experiences corroborate these findings. For example, natural oscillations of ~ 1.2 Hz appearing due to interactions among synchronous generators and an inverter-interfaced battery system in Cordova, Alaska, have been mitigated by tuning the inverter synthetic inertia parameters [18]. High-frequency subsynchronous oscillations driven by IBRs in Kauai Island, Hawaii, have been mitigated by switching the control mode of an IBR from grid-following (GFL) to grid-forming (GFM). Modeling studies showed that the mitigation could also have been achieved by tuning the GFL inverter's droop or phase-locked loop (PLL) parameters [19]. However, most of the works cited in the discussion thus far are case studies focused on specific interconnections, making it challenging to draw general conclusions applicable to systems with different network topologies.

To address this gap, this paper formulates an analytical framework for identifying the factors influencing changes in inter-area modes due to IBR penetration. The work uses GFM inverters as their industry adoption is expected to grow in the coming years. Droop control is the baseline GFM technology today, and hence has been used in this work. However, the analysis can be easily extended to other GFM controls such as virtual synchronous machine (VSM) control.

There are two ways in which IBR proliferation in bulk power grids impacts inter-area oscillations. First, changes in power flow patterns will change the network operating

point, thereby affecting its modes [20], [21], [22]. Second, differences in the dynamic behavior of synchronous machines and IBRs (e.g., loss of rotational inertia, fast frequency response) will change the dynamic response at buses in response to disturbances, thus changing the state-space model and system modes. In this paper, changes due to the differences in dynamics between IBRs and rotating machines are examined. By studying the sensitivity of the state matrix eigenvalues to locational changes in inertia, three factors influencing mode behavior are identified. These factors are the location where synchronous machines are replaced by IBRs, as well as two GFM parameters: the active power-frequency droop, and the time constant of the active power measurement filter. The mathematical analysis uses a second-order approximation for representing machine dynamics, and the findings are validated using dynamic simulation results of the IEEE 39-bus system [23] and the 2031 Western Electricity Coordinating Council (WECC) heavy-winter planning models.

The major contributions of this work are summarized as follows. First, an analytical expression is derived to identify factors influencing changes in inter-area mode properties due to growing IBR penetration in bulk power grids. Second, the analytic conclusions are validated using simulations on the full-topology IEEE 39-bus and 2031 WECC heavy-winter models. Simulation results show that IBR locations are the strongest factor affecting how modes change; but issues of concern, if observed, can be mitigated to some extent by tuning inverter parameters. In addition to corroborating earlier findings that inverter control parameters impact their damping contribution, this work shows how the extent of such impact is related to where an inverter is placed in a multi-machine bulk power network. In other words, identical changes in control parameters for inverters located in different points in the grid will result in unequal changes in inter-area mode properties (frequency, damping ratio, and shape). These insights will be instrumental for electric utilities, to assist understanding how inverter penetration in different parts of their footprints may affect system dynamics.

The remainder of the paper is organized as follows. Section II briefly revisits some preliminary concepts before delving further into eigenvalue sensitivity analysis in Section III. Insights derived in Section III are then validated using simulations conducted on the benchmark IEEE 39-bus and WECC 2031 heavy winter planning models in Section IV. Section V reiterates the key findings from this work and concludes this paper.

II. PRELIMINARIES

This section briefly reviews the fundamentals of inter-area oscillation modes in a power system, how they are related to the system's state-space model, and how a simplified state-space model for a power system with both IBR and synchronous machines may be constructed. The set-up described herein is utilized in the analysis presented in the subsequent sections of the paper.

Standard mathematical notations are used; calligraphic symbols represent sets, lower and upper case bold letters respectively indicate column vectors and matrices. All zero and all one vectors and matrices of appropriate size are denoted by $\mathbf{0}$ and $\mathbf{1}$ respectively; $\mathbf{I}$ is the identity matrix.

A. INTER-AREA OSCILLATION MODES

A brief overview of power system modes and their properties is provided here. The presented background is abridged and meant only for convenient reference. A fuller treatment of the material can be obtained from [3].

The state-space model of a power system can be expressed by linearizing around an operating point with the following equation:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{e}(t), \quad (1)$$

where $\mathbf{x}(t)$ is the state vector, $\mathbf{e}(t)$ is a vector describing control actions or perturbations applied to the system, $\mathbf{A}$ is the state matrix, and $\mathbf{B}$ is the input matrix. The dominant oscillatory modes of a system are related to the complex eigenvalues of the state matrix $\mathbf{A}$. Although the number of eigenvalues $\mathbf{A}$ has is determined by its order, only a subset of these manifest as dominant inter-area oscillation modes. The real components of the eigenvalues indicate the damping, and the imaginary components give the frequency of oscillation. A negative real component indicates damped oscillations, while a positive real part implies oscillations of increasing amplitude.

Let $\lambda_i = x_i + jy_i$ be the i -th eigenvalue of matrix $\mathbf{A}$, and $\mathbf{u}_i$ and $\mathbf{v}_i$ be the corresponding left and right eigenvectors. The oscillation frequency f_i and damping ratio ζ_i of the mode associated with λ_i are calculated as follows:

$$f_i = \frac{\text{Im}(\lambda_i)}{2\pi} \text{ Hz}, \quad (2)$$

$$\zeta_i = -\frac{\text{Re}(\lambda_i)}{|\lambda_i|} \times 100\%. \quad (3)$$

As eigenvalues appear in conjugate pairs, it is common to only examine those with positive frequencies. The entries of the right eigenvector indicate the corresponding mode shape, a measure of the modal observability of a state. Mode shapes indicate the relative activity of state variables when a particular mode is excited. The measure is described by two parameters- magnitude and angle. Mode shape magnitudes convey the degree of activity of the state variables in a mode, while angles give phase displacements of the state variables with regard to the mode. It is common to express the mode shapes of state variables relative to a common reference. The magnitude and phase angle of the mode shape (MS) observed between states x_j and x_k is given by:

$$|MS_{jk}| = \frac{|v_{i,j}|}{|v_{i,k}|}, \quad (4)$$

$$\angle MS_{jk} = \angle v_{i,j} - \angle v_{i,k}, \quad (5)$$

where $v_{i,j}$ and $v_{i,k}$ refer to the corresponding entries of the i -th right eigenvector of $\mathbf{A}$.

Left eigenvector entries denote the controllability of a mode from an actuator at a given location. Thus, the element $v_{i,k}$ measures the activity of the state variable x_k if the i -th mode is excited, while element $u_{i,k}$ weighs the contribution of this activity to the i -th mode.

B. POWER SYSTEMS MODEL

A power transmission network may be modeled as a connected undirected graph $\mathcal{G} := (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of vertices (buses) and edges (transmission lines), respectively. For ease of analysis, a network-reduced version of the system model can be constructed using Kron reduction where only the generator buses are retained [3], [24]. Consider a Kron-reduced network with n generator buses, where the dynamics at the i -th bus may be described as:

$$m_i \ddot{\theta}_i + d_i \dot{\theta}_i = p_{in,i} - p_{e,i}, \quad i \in \{1, 2, \dots, n\}. \quad (6)$$

Here, $p_{in,i}$ and $p_{e,i}$ respectively denote the input power and output electric power at the i -th bus. If the generator connected at bus i is a synchronous machine, then (6) is the classical swing equation where θ_i is the generator rotor angle, $m_i > 0$ is the rotational inertia, and $d_i > 0$ is the damping coefficient [3]. Moreover, $\frac{m_i}{d_i} = c_i$, where c_i is a constant dependent on machine design. On the other hand, if bus i is connected to a GFM inverter, then θ_i is the bus voltage phase angle. For a droop-based GFM, the inertia and damping terms become [25]:

$$m_i = \frac{t_{f,i}}{k_{p,i}}; \quad d_i = \frac{1}{k_{p,i}}, \quad (7)$$

where $k_{p,i}$ is its active power-frequency droop coefficient and $t_{f,i}$ is the time constant of the active power measurement low pass filter. Although the derivation of the above equivalent expressions is based on some simplifying assumptions, it facilitates the convenient application of stability analysis techniques developed using the swing equation to droop controllers. It must also be noted here that an ensemble of generators may be connected to each bus, and the inertia and damping terms in (6) will be lumped parametrizations of the aggregate behavior obtained.

In a power network, the power flow equation dictating the real power injection p_i at any bus i can be written as follows:

$$p_i = \sum_{j=1}^n |V_i||V_j|(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)) \quad (8)$$

Here, G_{ij} and B_{ij} respectively denote the real and imaginary parts of the corresponding entries in the admittance matrix of the system; V_i and V_j denote the voltage magnitudes at buses i and j assuming that they are connected, and θ_i and θ_j are the bus voltage angles. Three practical observations from high voltage electric transmission systems, as stated next, can be leveraged to simplify (8) and frame it as a linear problem.

Assumption 1: - Inductive lines: The resistance of transmission lines is much lower than their reactance. This leads

the real parts of admittance matrix entries to be much lower than the imaginary parts. Hence, the G_{ij} entries may be reasonably approximated as zero.

Assumption 2: - Unit Voltage magnitudes: In most operating conditions, the voltage magnitudes at transmission buses are maintained close to 1 per unit (typical operating ranges are 0.95-1.05 p.u.). Because the product of the V_i and V_j terms appears in (8), approximating these magnitude terms as 1 does not introduce high errors in power flow computations.

Assumption 3: - Small angle approximation: Under typical operating conditions, the voltage angle difference between two buses i and j connected by a transmission line is lower than 10-15 degrees, and it is very rare for this angular separation to exceed 30 degrees. Hence, the angular separation in transmission networks is considered 'small', where $\sin(\theta) \simeq \theta$ is an acceptable approximation.

Combining the assumptions stated above, the power flow equation in (8) may be stated as:

$$p_{e,i} = \sum_{j=1}^n b_{ij}(\theta_i - \theta_j), \quad (9)$$

where $b_{ij} \geq 0$ is the susceptance between nodes i and j if they are connected. The interested reader may refer [26] for a detailed derivation of power flow equations and their popular approximations. Using (6) and (9), the state-space representation of the transmission network can be expressed as [27]:

$$\begin{bmatrix} \dot{\theta} \\ \dot{\omega} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{L} & -\mathbf{M}^{-1}\mathbf{D} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \theta \\ \omega \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1} \end{bmatrix}}_{\mathbf{B}} \underbrace{\mathbf{p}_{in}}_{\mathbf{e}} \quad (10)$$

Here, the state variables $\theta \in \mathcal{R}^n$ and $\omega \in \mathcal{R}^n$ are the stacked vectors of angles and frequencies, and $\mathbf{p}_{in} \in \mathcal{R}^n$ is the net power input. The above quantities are all expressed as deviations from their nominal values. The matrices $\mathbf{M} = \text{diag}\{m_i\}$ and $\mathbf{D} = \text{diag}\{d_i\}$ are $(n \times n)$ diagonal matrices made up of the inertia and damping terms. The network Laplacian matrix $\mathbf{L} \in \mathcal{R}^{n \times n}$ has diagonal elements $l_{ii} = \sum_{j=1, j \neq i}^n b_{ij}$, and off-diagonal entries $l_{ij} = -b_{ij}$.

III. EIGENVALUE SENSITIVITY

Using the state-space model setup described in Section II, this section examines the sensitivity of the state matrix eigenvalues to locational changes in generator parameters, thereby identifying the major factors influencing changes in mode properties when synchronous machines at network nodes are replaced by IBRs. Eigenvalue sensitivity analysis studies have a well-established presence in power systems literature. In prior work, expressions for modal variations due to active/reactive power modulation at different points in the network have been derived to inform damping controller placements and power system stabilizer (PSS) and HVDC control strategies [21], [28], [29], [30]. In these works, the assumption is that these elements will be tuned to improve the damping of specific modes. The analysis

presented in this work borrows techniques from this rich prior literature and interprets the results in the context of mode changes when rotational inertia is displaced in parts of the grid. The objective of this work is not to derive the exact mathematical expression for changes in eigenvalue when an IBR replaces a synchronous machine, but rather to identify the major factors that dictate this change. Insights thus gained will be useful to bulk power system operators in understanding where within their footprint the loss of rotational inertia can lead to significant changes in inter-area modes.

One question that may arise here is: how realistic is it to assume that a retired synchronous machine will be replaced by an IBR with an equal power injection at the same location? While such one-to-one displacements in future grids are not expected, the experience of grid operators in the US suggests that large generator interconnection requests are being received in the vicinity of coal plants earmarked for retirement [31]. This may be because the proximity to pre-existing transmission infrastructure minimizes development costs, providing a lucrative option for renewable energy developers. This industry experience justifies the synchronous machine displacement scenarios studied in this paper.

A. SYNCHRONOUS MACHINE DISPLACEMENT

Replacing a synchronous machine from bus j , i.e. the loss of its rotational inertia can be viewed as changing the corresponding m_j terms in the state matrix of (10), which in turn impacts its eigenvalues. The sensitivity of the i -th eigenvalue λ_i with respect to (w.r.t.) a parameter α can be computed as [21], [28]:

$$\frac{\partial \lambda_i}{\partial \alpha} = \mathbf{u}_i^T \frac{\partial \mathbf{A}}{\partial \alpha} \mathbf{v}_i \quad (11)$$

In the current work, we want to determine how the state matrix eigenvalues will change if a synchronous generator at bus j is replaced by an IBR, i.e., m_j changes from m_{SM} (inertia of the displaced synchronous machine) to m_{IBR} (inertia of the IBR introduced into the network). Thus, we look at the eigenvalue sensitivity w.r.t. m_j next. Unlike IBRs, the inertia constant of a synchronous machine cannot be tuned adaptively. Hence, varying inertia at bus j must be interpreted as swapping the machine at bus j with another machine that has a different inertia constant, adding or retiring units in existing generation plants, etc.

A change in rotational inertia at bus j only affects state matrix entries in the row corresponding to the j -th bus. In the state matrix of (10), this would be row $(n + j)$. The entries of the $(n + j)$ -th row can be expanded as:

$$A_{n+j} = - \left[-\frac{b_{j1}}{m_j} \cdots \sum_{k=1, k \neq j}^n \frac{b_{jk}}{m_j} \cdots -\frac{b_{jn}}{m_j} \mid 0 \cdots c_j \cdots 0 \right] \quad (12)$$

Now, computing the partial derivative of $\mathbf{A}$ w.r.t. m_j yields non-zero entries only for the $(n + j)$ -th row. The partial derivative becomes:

$$\frac{\partial \mathbf{A}}{\partial m_j} = \begin{bmatrix} 0 & \cdots & 0 & | & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & | & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & | & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & | & \vdots & \ddots & \vdots \\ \frac{b_{j1}}{m_j^2} \cdots - \sum_{k=1}^n \frac{b_{jk}}{m_j^2} \cdots \frac{b_{jn}}{m_j^2} & | & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & | & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & | & 0 & \cdots & 0 \end{bmatrix} \quad (13)$$

Substituting (13) into (11), it is seen that:

$$\begin{aligned} \frac{\partial \lambda_i}{\partial m_j} &= -u_{i,n+j} \left(v_{i,1} \frac{b_{j1}}{m_j^2} \cdots - v_{i,j} \sum_{k=1}^n \frac{b_{jk}}{m_j^2} + \cdots v_{i,n} \frac{b_{jn}}{m_j^2} \right) \\ &= -\frac{u_{i,n+j} v_{i,j}}{m_j^2} \sum_{k=1}^n \left(\frac{v_{i,k}}{v_{i,j}} - 1 \right) b_{jk} \end{aligned} \quad (14)$$

Close inspection of (14) shows that the change in state matrix eigenvalues due to a change in inertia at a network node depends on- a) the amount of inertia displaced, and b) where the change occurred. Several terms in (14) relate to the node where inertia is varied - $v_{i,j}$ is the i -th mode shape associated with θ_j before the generator displacement, $\frac{v_{i,k}}{v_{i,j}}$ indicates the relative mode shapes of the rotor angles/bus voltage phase angles at neighboring machines, and b_{jk} is the susceptance of the lines connecting bus j to its neighbors. The term $u_{i,n+j}$ is the left eigenvector entry corresponding to ω_j , that expresses the controllability of the i -th mode in the state variable w_j . Thus, mode changes will be dictated by the mode observability and controllability in the network location where inertia is displaced, as well as how strongly the affected node is connected to the network.

If the displaced generator had high participation in a local oscillation mode, then the change in mode properties will primarily impact the neighboring buses. On the other hand, if it participated in an inter-area mode, then changes would be distributed across all machines participating in that mode. The result derived here is applicable irrespective of the configuration of the inverter that replaces the retired synchronous machine. If the inverter uses droop-based GFM control, then the corresponding change in inertia will be $\Delta m_j = m_{SM} - m_{IBR} = m_{SM} - \frac{t_{f,j}}{k_{p,j}}$, leading to the conclusion that the inverter parameters (droop coefficient and active power measurement filter time constant) will also affect the observed changes.

B. INVERTER DROOP-CONTROL PARAMETER

When a droop-controlled GFM inverter is introduced into a bulk power grid, its droop coefficient dictates its dynamics, and hence any subsequent impact on inter-area mode properties. By tuning the droop coefficient, modal properties can be affected. The sensitivity of the i -th eigenvalue to the tuning of droop parameter of a GFM inverter connected to the j -th bus can be determined using the chain rule as follows:

$$\begin{aligned} \frac{\partial \lambda_i}{\partial k_{p,j}} &= \frac{\partial \lambda_i}{\partial m_j} \times \frac{\partial m_j}{\partial k_{p,j}} \\ &= \frac{\partial \lambda_i}{\partial m_j} \times \left(-\frac{t_{f,j}}{k_{p,j}^2} \right) \\ &= -u_{i,n+j} v_{i,j} \times \frac{k_{p,j}^2}{t_{f,j}^2} \sum_{k=1}^n \left(\frac{v_{i,k}}{v_{i,j}} - 1 \right) b_{jk} \times \left(-\frac{t_{f,j}}{k_{p,j}^2} \right) \\ &= \frac{u_{i,n+j} v_{i,j}}{t_{f,j}} \sum_{k=1}^n \left(\frac{v_{i,k}}{v_{i,j}} - 1 \right) b_{jk} \end{aligned} \quad (15)$$

Thus, two GFM inverter parameters can be tuned to adjust modal properties - the active power-frequency droop coefficient $k_{p,j}$ and time constant of the active power measurement low pass filter $t_{f,j}$. The controllability term $u_{i,n+j}$ appears in (15), validating the intuition that parameter tuning will have higher impact at locations where mode controllability is high.

IV. SIMULATION RESULTS

The results in Section III have been derived using a second-order approximation of the linearized swing model of power system dynamics. In this section, these conclusions are validated using simulations conducted on the full dynamic models of two systems - the IEEE 39-bus benchmark [23] and the 2031 heavy-winter planning model of the US Western Interconnection. Transmission lines are represented using the pi-model. All simulations are carried out using Siemens PTI PSS/E transient simulation tool.

A. IEEE 39-BUS SYSTEM

There are ten conventional generators in the benchmark IEEE 39-bus system. For simulations performed in this work, the sixth-order round rotor generator model (GENROU) was used to represent the machine dynamics of generators 30-38. The GENROU model includes sub-transient, transient, and synchronous parameters. Generators 30-38 are equipped with governors such as GAST, TGOV1, IEEEESGO, and IEEEG1 to provide primary frequency control. For generator 39, which serves as the swing bus and represents the system's connection to the larger electric grid, the constant internal voltage generator model (GENCLS) is used to depict its dynamics. This is a second-order model. Unlike the other generators, generator 39 is not equipped with a governor.

To investigate the impact of IBR integration at different buses, synchronous machines in the benchmark IEEE 39-bus system (see Fig. 1) were replaced one at a time by droop-

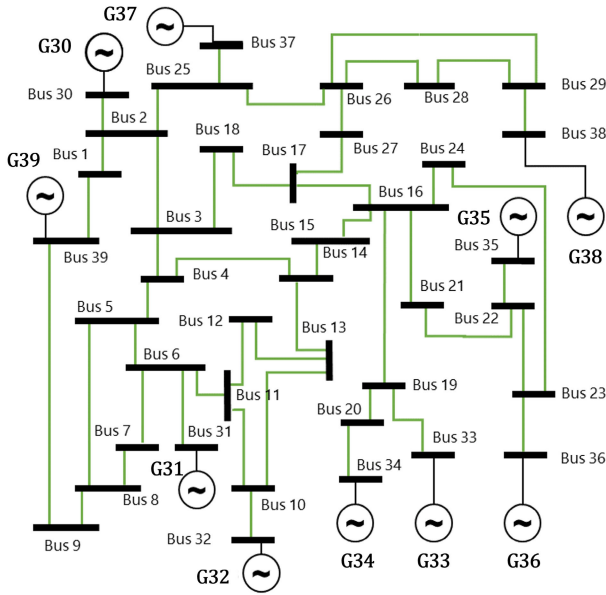


FIGURE 1. IEEE 39-bus benchmark system, adapted from [23].

controlled GFM inverters. The consequent changes in the inter-area mode properties from the baseline case were then studied. IBRs in the modified cases injected the same amount of power as the synchronous generators they replaced; thus keeping the initial power flow solution unchanged. The total in-service load in the base case was 6097.5 MW. Modes were excited by simulating a 600 MW dynamic brake insertion for 0.5 seconds at bus 31, and the post-disturbance signals were analyzed using the multi-channel Prony analysis method to estimate mode properties [32]. The Prony method is a well-established approach for mode estimation using measurements, and expresses the post-disturbance “free response” system output as a linear combination of damped sinusoids. Using the data-driven mode estimation technique ensures that changes in mode properties visible from the full system dynamic model, and hence, any deviation from the second-order approximation used in the derivations of Section III, are adequately captured.

1) INVERTER MODEL

The GFM inverter model REGFM_A1 used in this work employs single-loop droop-based control [33]. The inverter is modeled as a voltage source behind a coupling impedance, and the droop controller dictates the inverter’s internal voltage magnitude and angular frequency. A fault current limiting function that limits the output current during short-circuit conditions is also included [34].

While it is true that inverter control strategies vary across manufacturers, and only black-box models of their dynamic behavior are available to system operators, using generic models in system-level studies ensures computational tractability. The REGFM_A1 model, developed by The Universal Interoperability for Grid-forming inverters (UNIFI) Consortium and benchmarked against prevalent commercial

implementations, has recently been approved by the WECC Modeling and Validation Subcommittee (MVS) for use in initial studies investigating the impact of IBR penetration on bulk power grids and is also available on popular simulation softwares such as PSS/E and PowerWorld [35]. Therefore, using this model ensures reproducibility in future studies by the authors and other researchers. Default parameters from the REGFM_A1 model were used with 1% P-f droop and $t_f = 0.01s$.

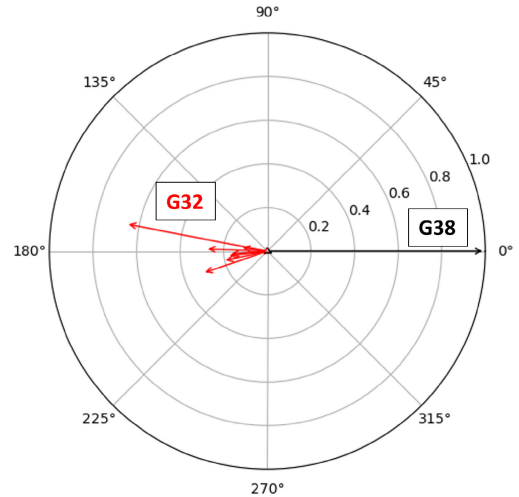


FIGURE 2. Mode shape plot for the 0.87 Hz mode in the unmodified IEEE 39-bus system.

2) IMPACT OF INVERTER LOCATION

In the original IEEE 39-bus system, the dynamic brake insertion simulated at Bus 31 excited an inter-area mode with frequency ~ 0.87 Hz and DR 7.46%. The corresponding mode shape observed at different generators is shown in the compass plot of Fig. 2. In the figure, the size and direction of the arrows respectively indicate the mode shape magnitude and angle observed at a location. Two generator groups are observed, denoted by red and black arrows in Fig. 2; generators G38 and G37 swing against the others in the system. For convenient reference, the mode shape values are noted in Table 1.

TABLE 1. Mode shape estimates for the 0.87 Hz mode in the IEEE 39-bus system.

Bus	Magnitude	Angle
30	0.0253	-141.3°
31	0.2867	177.96°
32	0.6616	169.09°
33	0.179	-175.14°
34	0.1872	-173.12°
35	0.3157	-161.09°
36	0.2083	-167.55°
37	0.0405	-26.72°
38	1	0°
39	0.1281	171.79°

Next, the synchronous generators in the base case were replaced one at a time by GFM inverters with equivalent injections, and the resultant changes in the properties of the 0.87 Hz mode were tracked. The IBR capacities were set such that 10% headroom was available for responding to disturbances. Two generators were not replaced in the simulated cases - a) G31 as the studied disturbance was placed at bus 31, and b) G39 as bus 39 is the swing bus in the system representing its point of connection to the larger electric interconnection, and G39 is the lumped representation of the larger grid modeled as a generator with inertia much higher than the other machines in the system. Because the inertia constants of the generators replaced in the simulated cases were similar, the observed changes in mode properties were primarily driven by the location where the change occurred. It must be mentioned here that the 39-bus system comprises additional oscillation modes, but the 0.87 Hz mode has been studied here because it was easily observable from the disturbance simulated in this work.

For each of the simulated synchronous machine replacement cases, the properties of the 0.87 Hz mode were estimated. In Fig. 3 and 4, the changes in mode property estimates are plotted against the mode shape magnitude observed at the IBR bus in the base case. It is evident from the increasing trend in the figures that when a synchronous machine is replaced by an IBR at a bus that has a high mode shape magnitude, larger changes in mode frequency and DR are observed, which is consistent with the conclusions from (14). The trend is not monotonous as the mode changes are also impacted by the susceptance of the lines connecting the IBR bus to its neighbors and the mode shapes of the neighboring buses. Although Fig. 4 shows that damping generally increases when IBRs are placed at buses with high mode shape magnitudes, it must be noted that this behavior also depends on inverter parameter tuning. A poorly tuned inverter may also decrease oscillation damping. This damping reduction may be greater if the inverter is placed at a bus with a high mode shape magnitude compared to a bus with a lower mode shape magnitude.

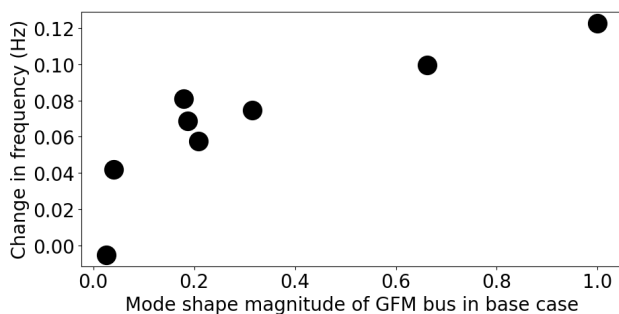


FIGURE 3. Change in mode frequency estimates with the mode shape magnitude of the IBR bus in the base case.

These conclusions can also be validated by examining the time-domain trajectories of the system bus frequencies for

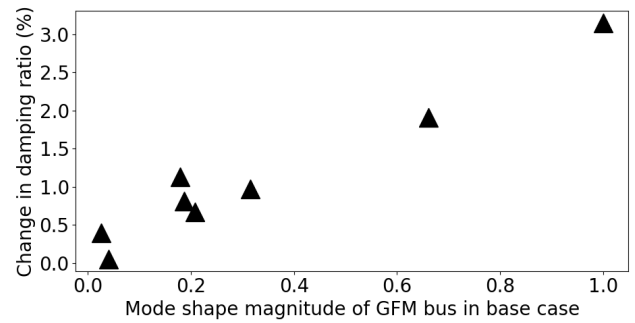


FIGURE 4. Change in DR estimates with the mode shape magnitude of the IBR bus in the base case.

the various simulated cases. As an illustrative example, Fig. 5 shows the frequencies obtained at Bus 32 when the GFM IBR is placed at different points in the network. The solid black trace shows the frequency trajectory in the base case. Note that when the IBR is placed at Bus 30 (light blue dot-dash line) which had a low mode shape magnitude in the base case, the oscillation frequency and time taken for them to decay do not change drastically from the base case. On the other hand, when the IBR is placed at bus 38 which had a higher mode shape magnitude (red dotted line), the oscillations decay faster and have a higher frequency, hence displaying significantly different behavior from the base case. When the inverter is at Bus 32, the observability of oscillations in Bus 32 frequency reduces greatly (solid green line), but oscillations continue to persist at other synchronous machines in the network as visible in Fig. 6.

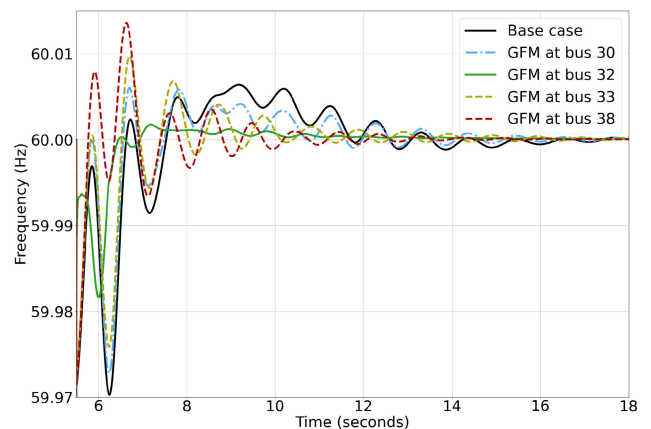


FIGURE 5. Post-disturbance ringdown frequency observed at bus 32 when the synchronous machine at different locations in the IEEE 39-bus system is replaced by a droopcontrolled GFM inverter. The simulated disturbance is a dynamic brake insertion at bus 31 between 5-5.5 seconds.

The results additionally illustrate that if IBRs displace synchronous machines, the mode shapes observed across the network are also affected. Fig. 7 shows the mode shape plot obtained for the case where the synchronous machine G32 is replaced by a GFM IBR. Note that the relative grouping among generators change; unlike the base case, generator G31 swings in phase with G37 and G38, and

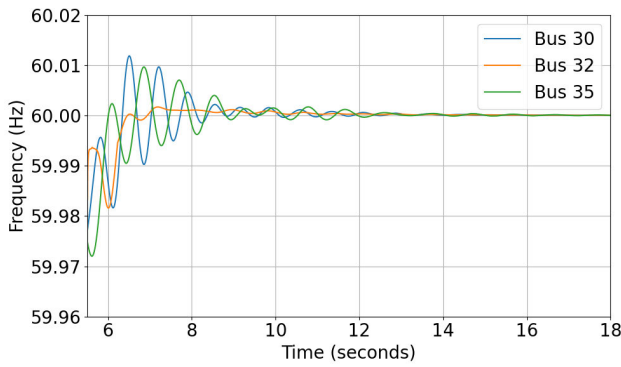


FIGURE 6. Post-disturbance ringdown frequency at generator buses of the IEEE 39-bus system when the generator G32 is replaced by a GFM IBR.

the angular difference among generators in the red group increases compared to the base case.

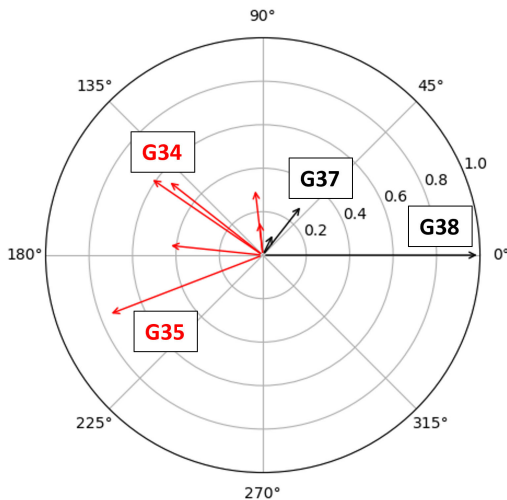


FIGURE 7. Mode shape estimates obtained when the generator G32 is replaced by a GFM IBR.

3) TUNING INVERTER PARAMETERS

As previously mentioned, changes in grid oscillatory behavior introduced by the displacement of synchronous machines/integration of IBRs can be mitigated to some extent by tuning inverter parameters. Using simulations on the IEEE 39-bus system, Fig. 8 and 9 show the variation in mode frequency and damping ratio estimated when different droop and t_f parameter values are used for the GFM inverter replacing the synchronous generator G36. The parameters are varied over typical value ranges prescribed: t_f from 0.01 to 0.1 s, and k_p between 0.01 and 0.05 p.u.. Results show that the mode appears to be better damped and has higher frequency when the droop settings is lower. Further, the total change in mode properties observed when the droop parameter is varied from 1% to 5% decreases with an increase in t_f . This is consistent with (15) where t_f appears in the denominator of the eigenvalue sensitivity expression.

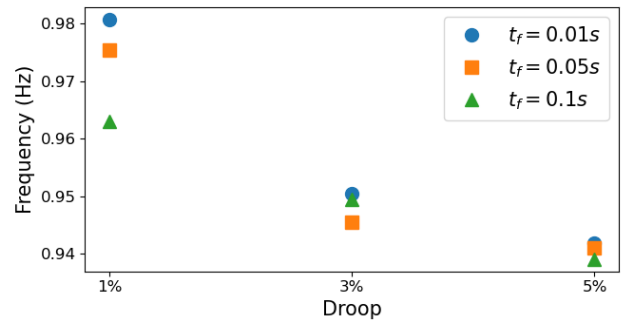


FIGURE 8. Variation in mode frequency estimates with inverter parameters.

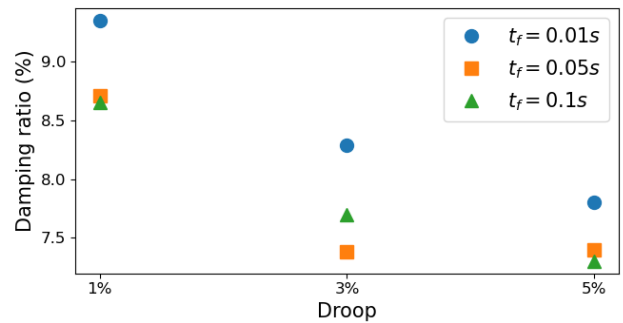


FIGURE 9. Variation in mode DR estimates with inverter parameters.

These results show that varying inverter parameters can change oscillation mode properties, but because the parameters are usually maintained within narrow specified ranges, the changes in observed in mode properties are also small when compared to the impact of inverter location.

TABLE 2. Characteristics of WI inter-area modes [6].

Mode	Frequency (Hz)	Shape
North-South A (NS-A)	0.2-0.3	Alberta vs. system
North-South B (NS-B)	0.35-0.45	Alberta vs. (northern U.S. + B.C.) vs. southern U.S.
East-West A (EW-A)	0.35-0.45	(Colorado + eastern Wyoming) vs. system
British Columbia A (BC-A)	0.5-0.72	Not well-understood
British Columbia B (BC-B)	0.6-0.72	Not well-understood
Montana (MT)	0.7-0.9	Montana vs. system

B. WECC 2031 HEAVY WINTER PLANNING MODEL

Following the August 1996 blackout caused by undamped inter-area oscillations in the WI [36], significant effort has been invested in examining the characteristics of its oscillation modes, through both model-based studies and synchrophasor measurement analyses. WECC has recently published a report detailing the present knowledge of WI inter-area modes, including their controllability, observability, and excitability [6]. There are several dominant WI

modes in the 0.1-1 Hz range, as summarized in Table 2. In earlier work, the authors have studied the impact of large-scale replacement of fossil fuel generators with GFM IBRs in the WI, and noted that the location where synchronous generators are clustered will dictate the properties of the low-frequency oscillation modes manifesting in the modified IBR-dominated system [9]. The analysis in [9] was focused on modes excitable from the northwestern part of WI where large hydropower generators are clustered. In contrast, this paper studies the impact of targeted generator replacements on the properties of the Montana mode, aimed at examining the validity of (14).

In the Montana (MT) mode, generators in Montana are observed to swing against the rest of the WI. The mode is well-damped ($DR \sim 10\%$), and is primarily driven by a large coal generation facility in Colstrip, MT, whose retirement is expected to significantly alter the mode characteristics. The mode frequency varies in the 0.7-0.9 Hz range depending on system conditions. Simulations presented in this section were performed using the WECC 2031 heavy-winter planning model, which has ~ 160 GW total generation, about 12.5% of which is contributed by inverter-interfaced renewable resources (solar, storage, and wind). To study the MT mode, a 1,200 MW dynamic braking resistor insertion was simulated at the Colstrip bus for a duration of 0.5 s, and the post-disturbance ringdown response was analyzed using Prony's method. Disturbances at Colstrip are known to excite the MT mode, facilitating studies [6].

Two cases were simulated in addition to the base WECC planning model. In Case I, two hydro-power units with a combined active power output of ~ 130 MW (combined maximum capacity: 141 MW) in Yellowtail, Montana, were replaced by GFM inverters, and in Case II, two hydro-power units with a combined active power output of ~ 155 MW (combined maximum capacity: 180 MW) in Dalles, Oregon, were replaced by inverters. The same GFM model and parameters as described in Section IV-A1 were utilized. As previously discussed, in the MT mode, generators in the MT region are observed to swing against the rest of the WI, and the mode shape magnitude at Yellowtail is higher than that at Dalles [6]. As a result, Case I results in a larger change in the MT mode characteristics than Case II, as shown in Table 3. Since the amount of synchronous generation and size of machines replaced was very small ($< 1\%$ of total load), drastic changes in mode properties or transient behavior were not expected. However, replacing the Yellowtail generators with IBRs impacts the system's dynamic response in Case 1, while the system behavior in the base case and Case 2 are practically indistinguishable. This is despite the fact that the size of machines displaced in Case 2 was larger than that in Case 1.

Fig. 10 and 11 show the post-disturbance frequency at two generator buses in Montana following the simulated brake insertion. Although it is challenging to discern small changes in mode properties from time-domain plots, one can still observe that the frequency curves for the base case and

Case 2 almost overlap. The frequency trajectories of the base case and Case 1 differ slightly.

TABLE 3. Montana mode property estimates obtained in the simulated cases.

Case	Frequency (Hz)	DR (%)
Base case	0.9161	11.71
Case I: GFM at Yellowtail	0.9125	15.25
Case II: GFM at Dalles	0.9177	11.68

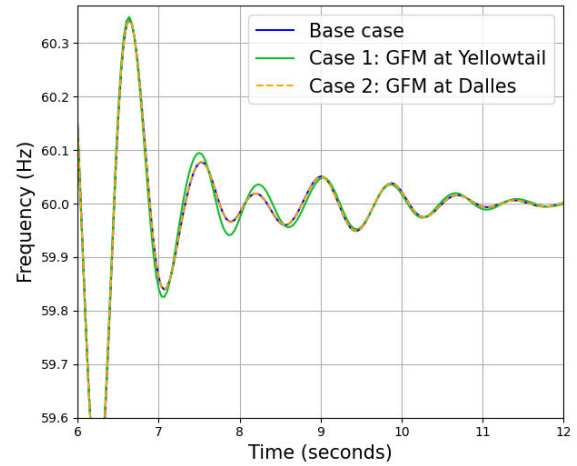


FIGURE 10. Post-disturbance frequency observed at a hydro-power generator bus in Montana. The dynamic brake insertion is simulated from 5 to 5.5 seconds.

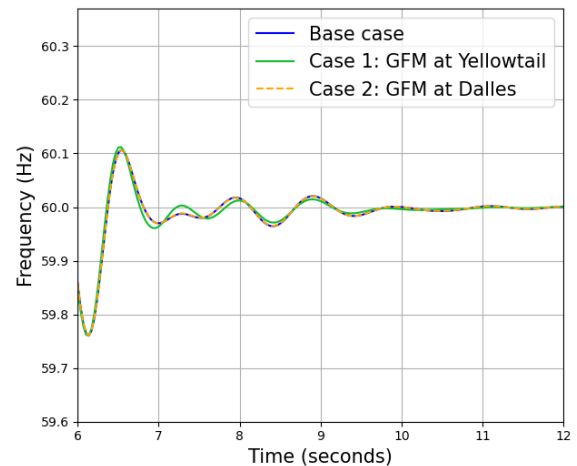


FIGURE 11. Post-disturbance frequency observed at a Type-4 wind power generator bus in Montana. The dynamic brake insertion is simulated from 5 to 5.5 seconds.

V. CONCLUSION

This paper investigates how the properties of oscillatory modes may change as IBR penetration increases in power systems. It is concluded that mode changes depend largely on the power network node where IBR penetration occurs and the amount of rotational inertia lost due to the displacement of synchronous machines. Two GFM inverter control

parameters- the active power-frequency droop coefficient, and the time constant of the active power measurement filter, are observed to impact mode properties. The addition of IBRs typically leads to an increase in oscillation mode frequency, and well-tuned GFM inverters can improve mode damping. Further, an identical change in inverter control parameters at different points in the grid results in unequal changes in mode properties.

These insights will be helpful for electric utilities in understanding how growing IBR penetration within their footprints may alter system dynamics. For example, they can identify areas where synchronous machine displacement by IBRs can significantly change existing mode properties, necessitating additional analysis during interconnection studies. Moreover, transmission nodes with high mode shape magnitudes and strong controllability can significantly impact oscillatory stability. Therefore, if IBRs are placed at these nodes, their control parameters should be carefully optimized.

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